Skeleton

1. The Solow-Swan Growth Model

The Shift: The Investment Curve $(s \cdot f(k))$ shifted **Vertically Upward**.

- **The Setup:**
  - **X-Axis:** Capital per worker (k).
  - **Y-Axis:** Output per worker (y).
  - **Curves:** The Production Function $y = f(k)$, the Investment Curve $s \cdot f(k)$, and the Depreciation Line $(n + \delta)k$.
- **The Shift (1965–1985):**
  - **Action:** The Savings Rate (s) did not just increase; it tripled.
  - **Shift:** The investment curve $s \cdot f(k)$ shifted **massively UPWARD**.
  - **Result:** The intersection with the depreciation line (the Steady State, k^*) moved dramatically to the **RIGHT**.
- **Exhaustive Reason:**
  - **Forced Savings (CPF):** The Central Provident Fund (CPF) is the key mechanism. In 1955, the contribution rate was 10% (5% employer + 5% employee). By 1984, the government legislated this up to **50%** (25% + 25%).
  - **Effect:** This exogenously increased the parameter s in the Solow equation. Singapore became a "capital accumulation machine," forcing the economy to a higher steady-state income not by efficiency, but by brute-force saving.
  - **Side Effect (Dynamic Inefficiency):** By the late 80s, economists argued Singapore had shifted the curve *too far* up, accumulating capital beyond the "Golden Rule" level (where saving more actually reduces consumption).

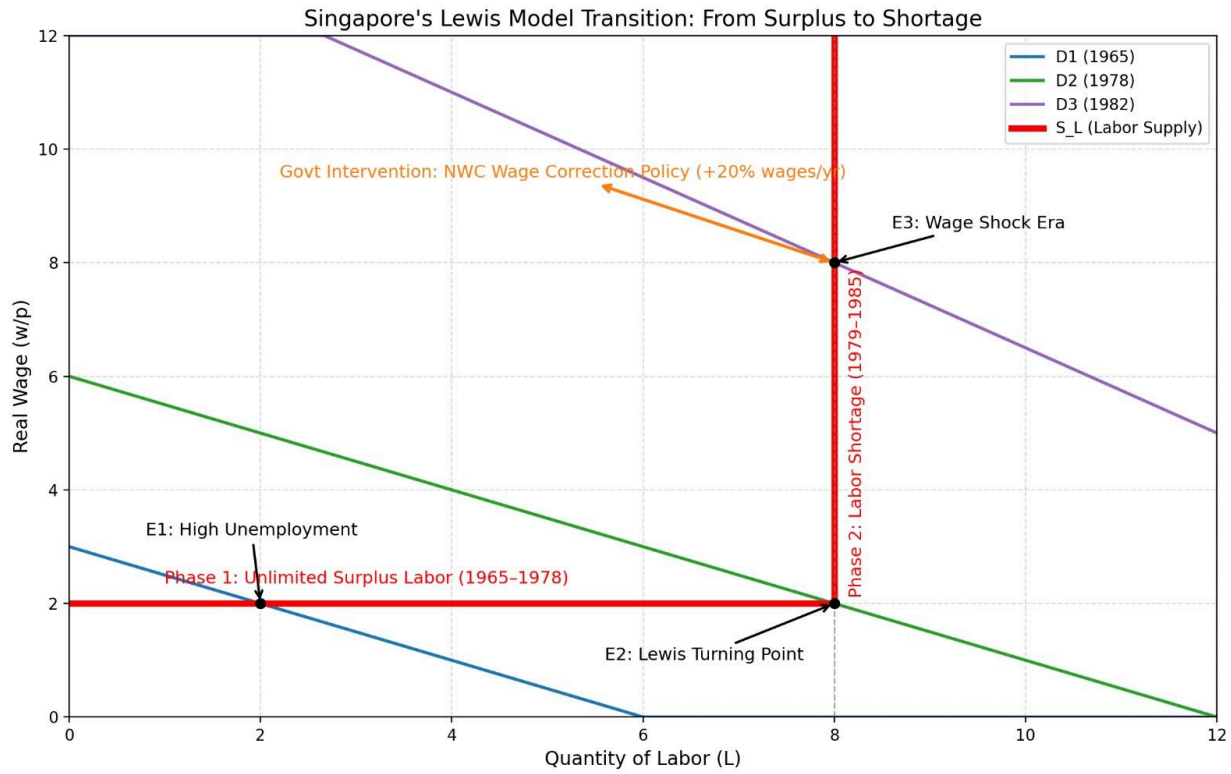
2. The Lewis Dual-Sector Model

The Shift: Movement **Rightward along** the Horizontal Supply Curve, then a sudden **Vertical jump** to the Inelastic Curve.

- **The Setup:**
  - **X-Axis:** Quantity of Labor (L).
  - **Y-Axis:** Real Wage (w/p).
  - **Curves:** Labor Supply (S_L) is "L-shaped" (flat at first, then vertical). Labor Demand (D_L) is downward sloping (VMP_L).
- **Phase 1 (1965–1978): The Horizontal Slide**
  - **Shift:** As EDB attracted MNCs, Labor Demand (D_L) shifted **RIGHT**.
  - **Result:** Because Singapore had "surplus labor" (unemployment ~10%), the equilibrium moved **RIGHT** along the flat part of the Supply curve.
  - **Outcome:** Employment increased ($L \uparrow$) but Wages stayed constant ($w \leftrightarrow$). This kept Singapore attractive to low-cost manufacturing.
- **Phase 2 (1979–1981): The Vertical Leap (The Turning Point)**
  - **Shift:** The economy hit the "Lewis Turning Point" where the Supply curve becomes vertical (inelastic).

- **Policy Intervention (The Shift):** The government imposed the "Wage Correction Policy."
- **Mechanism:** This was effectively a **Price Floor** set *above* the equilibrium.
- **Result:** This forced a movement **UP** along the Labor Demand curve.
- **Exhaustive Reason:** By artificially raising the price of labor, the government forced firms to increase the Marginal Product of Labor (MP_L) to justify the cost. Firms that couldn't (low-tech textiles) shut down. Firms that could (disk drives) automated.

Lewis



Yes, this graph **perfectly fits the concept** of the Lewis Dual Sector Model as applied to Singapore. It visualizes the exact "Singapore Story" we discussed, rather than just a generic textbook version.

Here is a breakdown of why this graph is accurate:

1. The "L-Shaped" Red Line (Labor Supply)

- **The Flat Part (0 to 8):** This represents the **1965–1978** era. The line is flat because Singapore had "Unlimited Surplus Labor." Even as demand grew (moving from $D1$ to $D2$), wages (Y-axis) stayed stuck at $2. This accurately shows how early industrialization created jobs without raising labor costs.
- **The Vertical Part (8 upwards):** This represents the post-1978 era. The line goes straight up because Singapore **ran out of workers**. The supply became "inelastic"—you cannot hire more people no matter what you pay.

2. The "Kink" at Point E2 (The Turning Point)

- The intersection at **E2** marks the **Lewis Turning Point** (around 1978). This is the exact moment Singapore hit full employment.
- The graph correctly shows that before this point, the curve was flat; after this point, it is vertical.

3. The Massive Jump to E3 (The Wage Shock)

- This is the most critical part for Singapore's context. In a normal free market, wages might creep up slowly.
- The graph shows a sharp jump from **E2 ($2)** to **E3 ($8)**. This large gap represents the **"Wage Correction Policy" (1979–1981)**.
- The annotation *"Govt Intervention: NWC Wage Correction Policy"* correctly identifies that this wasn't just market forces—it was a deliberate policy shock to force the economy up the value chain.

Verdict: This is an excellent, historically accurate representation of how Singapore manipulated the Lewis Model to force its industrial revolution.

Explanation

Here is the breakdown of the **Lewis Dual Sector Model** for Singapore, starting with a simple analogy for a 15-year-old and ending with the formal economic shifts.

Part 1: The 15-Year-Old Explanation

"The Overflowing Waiting Room"

Imagine Singapore in the 1960s as a factory with a **huge waiting room** outside.

- **The "Unlimited" Phase (1965–1978):**
  - There are 1,000 people in the waiting room begging for a job.
  - The Boss (MNCs) says: "I will pay you $2 an hour. Take it or leave it."
  - Because the waiting room is full, everyone takes the $2.
  - **Result:** The factory grows bigger and hires more people, but wages stay flat at $2. The Boss makes huge profits because labor is so cheap.
- **The "Turning Point" (Late 1970s):**
  - Suddenly, the waiting room is **empty**. Everyone has a job.
  - The Boss wants to hire 100 more people to expand.
  - He goes outside and shouts: "Who wants a job for $2?" **Silence.**
  - To get workers, he has to steal them from other factories. He shouts: "Okay, I'll pay $5!"
  - **Result:** Wages suddenly shoot up. The "cheap labor" era is over.
- **The Singapore Twist (The "Wage Shock"):**
  - The Government saw the waiting room was empty but didn't want the Boss to just hire cheap foreign workers to refill it.
  - So, the Government told the Boss: "You must pay $10 now. If you can't afford it, buy a robot or close down."
  - This forced the factory to upgrade from making T-shirts (cheap labor) to making Computer Chips (high skill).

Part 2: The Formal Economic Explanation

The Lewis Model explains how a developing economy transitions from a **Traditional Sector** (Subsistence/Farm) to a **Modern Sector** (Capitalist/Factory).

1. Phase 1: Unlimited Supply of Labor (The Horizontal Curve)

- **Context:** 1965–1975. Unemployment was >10%.
- **The Mechanism:** The "Traditional Sector" (kampungs, petty trade) had **Surplus Labor**. People were working, but their **Marginal Productivity was Zero** (if they left, output wouldn't drop).
- **The Graph:** The Labor Supply Curve (S_L) is **Horizontal (Perfectly Elastic)**.
- **Singapore's Growth:** As Foreign Direct Investment (FDI) flowed in, the **Labor Demand Curve (D_L) shifted RIGHT**.
    - Because Supply was horizontal, employment (L) increased massively, but Real Wages (w) remained constant. This fueled high profits and reinvestment.

2. The Lewis Turning Point (The Kink)

- **Context:** Late 1970s. Unemployment hit ~3% (Full Employment).
- **The Mechanism:** The "surplus labor" was exhausted. The Transfer of labor from traditional to modern sectors was complete.
- **The Graph:** The Supply Curve hits a "kink" and becomes **Vertical (Inelastic)**.
- **The Consequence:** Any further increase in Demand (D_L shifting Right) would no longer increase employment (L). Instead, it would purely drive up Wages (w), causing **Wage-Push Inflation**.

3. Phase 2: The "Wage Correction" Policy (The Vertical Shift)

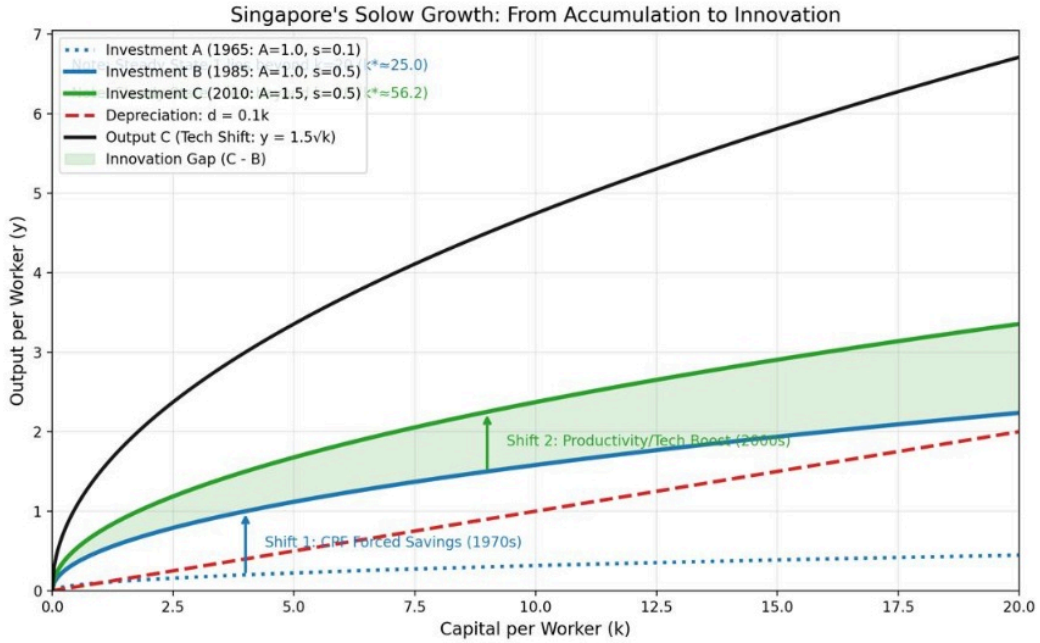
- **Context:** 1979–1981. The "Second Industrial Revolution."
- **The Policy:** The Singapore government realized that if they didn't intervene, firms would just import low-skilled foreign labor to keep the Supply Curve horizontal (keeping wages low).
- **The Intervention:** The National Wages Council (NWC) recommended **wage increases of ~20% per year** for 3 years.
- **The Graph Effect:** This effectively set a **Price Floor** above the market equilibrium.
- **The Result:** This destroyed the profitability of labor-intensive firms (textiles). They shut down or relocated to Malaysia/Indonesia. Capital-intensive firms (disk drives, petrochemicals) survived by substituting Labor (L) for Capital (K).

Part 3: The Shifts in the Graph (Step-by-Step)

Imagine a graph with **Quantity of Labor (L)** on the X-axis and **Real Wage (w)** on the Y-axis.

Phase	Curve Action	Direction of Shift	Economic Meaning
1. The Launch	**Demand Curve (D_L)**	**Shift RIGHT** ($\rightarrow$)	MNCs enter Singapore. They need workers.
2. The Expansion	**Equilibrium Point**	**Move RIGHT** ($\rightarrow$)	We move *along* the flat Supply curve. Jobs $\uparrow$, Wages $\leftrightarrow$ (Flat).
3. The Turning Point	**Supply Curve (S_L)**	**Become VERTICAL** ($\uparrow$)	We ran out of workers. The curve is no longer flat; it goes straight up.
4. The Squeeze	**Equilibrium Point**	**Move UP** ($\uparrow$)	Demand keeps shifting Right, but now it climbs the vertical wall. Wages explode.
5. The Intervention	**Wage Floor**	**Shift UP** ($\uparrow$)	Gov't forces wages *even higher* than the market. Low-value firms die.

Solow



Based on the context of our conversation, I will focus on the **"Singapore's Solow Growth"** graph (the second and third images you uploaded).

Verdict: Yes, this graph is **historically and theoretically accurate**. It is an excellent stylized representation of Singapore's unique two-stage economic journey. It correctly isolates the two main "engines" Singapore used to grow: first by brute-force saving (CPF), and second by innovation (Smart Nation).

Here is the intricate, step-by-step explanation of exactly what this graph is telling you.

1. The Setup (The Axes)

- **X-Axis (Capital per Worker k):** This represents the "hardware" of the economy—machines, factories, computers, and infrastructure available to each worker. Moving right means the country is getting more industrialized.
- **Y-Axis (Output per Worker y):** This represents GDP or Income. Moving up means the country is getting richer.

2. The Baseline: 1965 (The Dotted Blue Line)

- **Label:** Investment A (1965: A=1.0, s=0.1)
- **What it shows:** This is Singapore at independence.
    - **$s=0.1$:** The savings rate is very low (10%). People are poor and spend almost everything just to survive; they have little left to invest in new machines.
    - **$A=1.0$:** The technology level is basic (standard manual labor).

- **The Reality:** The dotted line is very flat. It intersects the Red Depreciation line very early (close to 0). This means the economy is stuck in a "poverty trap"—it cannot grow because it cannot save enough to replace the machines that wear out.

3. Shift 1: The "CPF Miracle" (The Blue Arrow)

- **The Shift:** From **Dotted Blue** $\rightarrow$ **Solid Blue Curve**.
- **Label:** Shift 1: CPF Forced Savings (1970s)
- **The Mechanism:** Notice that the Technology parameter (A) stays at **1.0**. We did *not* get smarter. Instead, the Savings parameter (s) jumps from **0.1 to 0.5**.
- **The Policy Story:** This visualizes the **Central Provident Fund (CPF)** policy. The government forced every worker to save huge portions of their salary (up to 50%).
- **The Visual Result:** The Investment curve rotates **vertically upward**.
- **The Economic Meaning:** Suddenly, Singapore had massive amounts of cash to build airports (Changi), ports, and factories. We moved to a "Higher Steady State" not by innovation, but by brute-force accumulation. This is the era of "perspiration, not inspiration."

4. Shift 2: The "Tech Boost" (The Green Arrow)

- **The Shift:** From **Solid Blue** $\rightarrow$ **Solid Green Curve**.
- **Label:** Shift 2: Productivity/Tech Boost (2000s)
- **The Mechanism:** This is the critical change.
    - The Savings rate (s) stays at **0.5** (we can't save any more than we already are).
    - The **Technology parameter (A) jumps from 1.0 to 1.5**.
- **The Black Line (Output C):** Look at the top solid black line. This is the *Production Function*. It shifts up because of the new technology ($A=1.5$). Because output (y) goes up, and we save a percentage of output, the Investment curve (Green) is pulled up with it.
- **The Policy Story:** This represents the post-2000s era: Biopolis, Fusionopolis, R&D grants, and the "Smart Nation" initiative. Singapore realized it couldn't just build *more* factories; it had to invent *better* ones.

5. The "Innovation Gap" (The Shaded Green Area)

- **What it represents:** This is the **"Free Lunch."**
- **Intricate Detail:** Look at any point on the X-axis (e.g., $k=10$).
    - The **Blue Line** tells you how much we would invest if we stayed with old technology ($A=1.0$).
    - The **Green Line** tells you how much we invest now with new technology ($A=1.5$).
- **The Difference:** The shaded area is the extra wealth generated **purely by efficiency**. It represents the income Singapore earns today not because it works harder than in 1985, but because it works smarter.

6. The Red Dashed Line (Depreciation)

- **Label:** Depreciation: d = 0.1k
- **The Limit:** This line represents the "cost of maintenance." Machines rust, roads crack, and computers become obsolete.
- **The Crossing Point (Steady State):** Economic growth *stops* where the Investment curve crosses this Red line.
    - **1965 (Dotted Blue):** Crosses very early (Singapore stays poor).
    - **1985 (Solid Blue):** Crosses much later (Singapore becomes a developed manufacturing hub).
    - **2010 (Solid Green):** Crosses even further right (Singapore becomes a wealthy, high-tech nation).

Summary:

This graph perfectly captures the "Two Engines" of Singapore's history. **Engine 1 (Blue Shift)** was turning up the savings dial. **Engine 2 (Green Shift)** was upgrading the engine itself (Technology).

Final Pointers

Topic: How Trade and Globalization Played a Major Role in Making Singapore a Success

Slide Section 1: Major Trade & Globalization Policies

- **Aggressive Attraction of Foreign Direct Investment (FDI):**
  - Strategy relied on the Economic Development Board (EDB) to attract MNCs.
  - This inflow of FDI caused the Labor Demand Curve to shift to the RIGHT, driving employment.
- **Export-Oriented Industrialization:**
  - Singapore positioned itself as a low-cost manufacturing hub for global markets.
  - Wages were kept constant initially to ensure the country remained attractive to investors.
- **State-Driven Structural Transformation (Wage Correction):**
  - The government intervened via the National Wages Council (NWC) to force the economy up the value chain.
  - Policy deliberately destroyed the profitability of low-tech, labor-intensive firms (like textiles) to make way for capital-intensive ones.

Slide Section 2: The Solow-Swan Growth Model Data

- **Phase 1: The "Capital Accumulation" Machine (1965–1985)**
  - **Core Driver:** "Forced Savings" via the Central Provident Fund (CPF) mechanism.
  - **Specific Data Shift:** The savings rate (s) tripled, moving from 10% (1955) to 50% (1984).
  - **Graph Impact:** The Investment Curve (s * f(k)) shifted massively vertically UPWARD.
  - **Outcome:** The economy moved to a higher steady state through brute-force accumulation rather than efficiency.
  - **Constraint:** By the late 80s, this led to "Dynamic Inefficiency," where capital was accumulated beyond the "Golden Rule" level.
- **Phase 2: The "Innovation" & Tech Boost (2000s–Present)**
  - **Core Driver:** A shift from "perspiration" to "inspiration" (e.g., Biopolis, Smart Nation).
  - **Specific Data Shift:** The Technology parameter (A) increased from 1.0 to 1.5, while savings (s) remained steady at 0.5.
  - **Graph Impact:** The entire Production Function shifted UP because of the new technology.
  - **Outcome:** This created an "Innovation Gap" (Free Lunch)—extra wealth generated purely by efficiency.

Slide Section 3: The Lewis Dual-Sector Model Data

- **Phase 1: Unlimited Surplus Labor (1965–1978)**

    - **Initial State:** High unemployment (>10%) with surplus labor in the traditional sector.
    - **Curve Shape:** The Labor Supply Curve (S_L) was Horizontal (Perfectly Elastic).
    - **Wage Dynamic:** Wages stayed flat at $2 (E1 to E2) despite massive increases in employment.
    - **Economic Impact:** This kept Singapore attractive for low-cost manufacturing MNCs.
- **Phase 2: The Turning Point & Wage Shock (1979–1985)**
    - **The Turning Point:** Reached around 1978 when unemployment dropped to ~3% (Full Employment).
    - **Curve Shape:** The Labor Supply Curve became Vertical (Inelastic) at L=8 because the local workforce was exhausted.
    - **Policy Intervention:** The NWC "Wage Correction Policy" recommended wage increases of ~20% per year.
    - **Graph Impact:** A massive vertical jump in the equilibrium wage from $2 (E2) to $8 (E3).
    - **Outcome:** This acted as a "Price Floor," forcing firms to increase marginal productivity or shut down.